Mohan Teja Nandamudi | Data Engineer

VA, USA | +1 508 598 8940 | mohanteja.0117@gmail.com | LinkedIn

SUMMARY

Data Engineer with 6+ years building and optimizing enterprise data pipelines, cloud warehouses, and real-time ingestion frameworks across AWS, GCP, Snowflake, Spark, Kafka, Airflow, Python, and SQL environments. Delivered measurable gains in pipeline stability, reporting speed, cloud cost efficiency, and migration outcomes for high-volume finance and eCommerce platforms. Hands-on expertise in batch processing, streaming architecture, CI/CD automation, dimensional modeling, data governance, SLA-driven production support, and analytics enablement. Experience across Capital One, American Express, and Amazon aligned to modern US hiring demand.

TECHNICAL SKILLS

Languages & Scripting: Python, SQL, PySpark, Java, Shell Scripting
Data Engineering & Processing: Apache Spark, Spark SQL, PySpark, Kafka, Airflow, DBT, Databricks, Hadoop, Hive, CDC Pipelines, Batch Processing, Streaming Pipelines
Cloud Platforms: Amazon Web Services AWS, Google Cloud GCP, Microsoft Azure Azure
AWS Services: S3, Glue, EMR, Redshift, Athena, Lambda, Kinesis, IAM, ECS, CloudWatch, Step Functions
GCP Services: BigQuery, Composer, Dataflow, Dataproc, Cloud Storage, Cloud Functions, Monitoring
Databases & Warehouses: Snowflake, Redshift, BigQuery, SQL Server, PostgreSQL, MySQL, Teradata
DevOps & Automation: Jenkins, GitHub Actions, Docker, Terraform, CloudFormation, CI/CD Pipelines, Git
BI & Reporting: Power BI, Tableau, QuickSight
Architecture & Governance: ETL, ELT, Data Modeling, Star Schema, Dimensional Modeling, Data Quality, Metadata Controls, Access Governance, Performance Tuning, Cost Optimization, Production Support

PROFESSIONAL EXPERIENCE

Data Engineer **Aug 2025 - Present | McLean, VA**
Capital One

- Rebuilt delayed finance reporting pipelines on AWS and Snowflake, cutting refresh cycles forty-two percent and restoring on-time executive dashboards across critical monthly closes.
- Consolidated fragmented datasets into governed S3 lake architecture, reducing duplicate storage thirty-five percent and eliminating repeated analyst extracts across multiple departments companywide.
- Replaced manual deployments with Airflow, DBT, and CI/CD automation, shrinking release effort sixty percent while reducing avoidable production defects significantly enterprisewide.
- Launched Kafka and Spark streaming framework detecting transaction anomalies faster, enabling operations teams to respond before downstream customer-impacting issues escalated further.
- Tuned expensive Snowflake workloads through clustering, caching, and SQL rewrites, reducing runtime forty percent and lowering recurring monthly compute charges materially.
- Strengthened cloud controls using IAM, encryption, and lineage tracking, improving audit readiness for regulated datasets reviewed during internal compliance assessments.
- Delivered executive KPI dashboards replacing spreadsheet packs, reducing ad hoc reporting requests forty-five percent and improving leadership decision turnaround times significantly.

Data Engineer **Nov 2024 – Aug 2025 | Phoenix, AZ**
American Express

- Led Teradata exit program to BigQuery, retiring legacy workloads while improving elasticity, stability, and analytics speed across enterprise finance reporting environments.
- Built Python conversion engine translating thousands of BTEQ scripts into BigQuery SQL, removing months of manual remediation effort across migration teams.
- Designed Cloud Composer orchestration managing ingestion, reconciliation, validation, and transformation workflows with dependable scheduling across mission-critical production datasets daily.

- Converted aging SAS jobs into Python and PySpark pipelines, preserving business outputs while simplifying maintenance and future enhancement delivery cycles.
- Reduced BigQuery spend forty-five percent through partitioning, clustering, and materialized views while noticeably accelerating high-volume analyst query performance enterprisewide.
- Implemented proactive logging and alerting controls, helping support teams identify failures early and recover pipelines before SLA breaches occurred.
- Partnered with security leaders implementing CMEK and IAM standards, ensuring migrated data assets met governance expectations from day one.

Data Engineer **May 2022 - Nov 2024 | Dallas, TX**
Amazon

- Built high-volume ETL pipelines supporting inventory, fulfillment, customer, and operations datasets consumed by multiple analytics teams across fast-moving business functions daily.
- Developed Spark processing frameworks handling billions of records reliably, increasing throughput while sharply reducing recurring failures during overnight production schedules.
- Created dimensional marts that standardized metrics definitions, helping teams launch dashboards faster and avoid conflicting reports during leadership reviews.
- Implemented Airflow controls with retries, dependencies, and alerts, improving workflow reliability for critical production pipelines requiring uninterrupted execution.
- Introduced reconciliation checks and schema validation, catching data issues early before downstream analysts consumed inaccurate operational metrics widely.
- Automated environment provisioning through Terraform templates, reducing setup delays and minimizing drift issues across shared engineering deployments significantly.
- Partnered with analysts and scientists delivering trusted datasets for experiments, recommendations, and growth decisions tied to measurable business outcomes.

Data Engineer **Jan 2020 - Aug 2021 | India**
Genpact Solutions

- Built Spark pipelines transforming raw warehouse, API, and flat-file data into trusted datasets ready for recurring analytics and reporting consumption.
- Implemented Kafka streaming integrations converting live events into structured outputs supporting operational dashboards and near real-time business visibility needs.
- Reduced recurring batch delays thirty-five percent by tuning Spark memory, caching, and execution plans across unstable scheduled workloads.
- Migrated legacy on-prem pipelines onto AWS services, improving scalability, recoverability, and long-term operating efficiency across data operations significantly.
- Delivered Tableau and Power BI dashboards with row-level security protecting sensitive metrics while expanding stakeholder access responsibly enterprisewide.
- Supported CI/CD releases using Jenkins, Git, and containers, helping engineering teams ship updates faster with fewer deployment errors.
- Worked in Agile sprint teams delivering fixes and enhancements consistently while maintaining dependable service levels across supported applications.

EDUCATION

Masters of Science in Data Science **Dec 2022**
University of Massachusetts Dartmouth, Dartmouth, MA